

Youth Risk for Substance Abuse & Family Dynamics: A Study In Chile

IDC 6940 Capstone in Data Science
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Abstract—This case study investigates what combinations of risk factors predict the youth groups most likely to consume drugs, through the use of predictive modeling. To determine the combinations, data were acquired from the Servicio Nacional para la Prevención y Rehabilitación del Consumo de Drogas y Alcohol (SENDA), a department under the Chilean Interior Ministry. The data comes from an annual questionnaire given to students in the 8° básico y 4° medio in the Chilean school system (equivalent to 8th grade and 12th grade in the U.S. school system). The questions in the survey relate to the family dynamics at home, their history of drug consumption, and their academic performance in school. The dataset will have responses from 2001 to 2013 for the analysis. This analysis will include preliminary patterns within the region, such as demographic patterns and descriptive statistics, that will determine which factors can be implemented into the predictive modeling. The predictive modeling generates an understanding of how drug crimes and substance abuse go hand-in-hand in Chile, a heavily debated issue during the 2025 presidential elections. In addition, it provides insights into how Chile responds to substance treatment policies and tackles drug trafficking throughout the nation.

Index Terms—substance abuse, Chile, consumption, drug use, youth

I. INTRODUCTION

A. Overview

Chile is regarded as one of the most stable countries in Latin America; however, the country has recently been observing a rise in illicit drug trafficking and consumption. Despite maintaining a secure economy and democracy post-dictatorship of Augusto Pinochet, Chile is struggling with the rise of crime and drug trafficking [1]. The disturbance of a country, known for their unique distinction from the Latin American region, was heavily debated during the 2025 general elections, with many candidates focusing on security and immigration [2]. This has caused growing concerns among the general public, specifically how it has impacted Chilean youth and what it can mean for the future. While external social factors contribute to the rise of drug consumption among Chilean adolescents, it is important to note that family dynamics can impact youth and be a predictive factor for the rise of drug consumption [3]. Overall, the reason that drug consumption increases through trafficking solely ignores the fact that what happens at home can also be an influence.

B. Motivation

Substance use among the youth is important to note as an issue in Chilean society, as it reflects upon academic performance and behavioral patterns. While drug trafficking is increasing in the country, the societal attitudes towards drug use and family relationship dynamics can influence how an adolescent approaches the use of drugs and the possibility of abusing them. This ongoing research addresses predictive factors of Chilean drug consumers by considering the youth's interactions with their parents and peers. In addition, this research provides a glimpse into my master's program in Latin American & Caribbean Studies, concentrating on public health in the region.

C. Goals & Objectives

The primary goals of this research are to identify patterns among Chilean youth, specifically adolescents who are in the high school level in the Chilean education system (8th-12th grade equivalent) whether there is an influence of the family towards drugs and to identify which factors are influential for observing which groups are more likely to consume drugs through predictive modeling. The factors will be based on variables such as socioeconomic status, peer exposure, and family relationships to examine patterns across different groups of status in the country. By doing so, this case study aims to provide prevention programs implemented in the country with more guidance on consumption-reduction initiatives.

D. Research Questions

I am currently exploring the following research questions: (1) What is the relationship between parental attitudes and youth behavior? (2) What combinations of risk factors predict the groups most likely to consume drugs? (3) Do prevention programs show improvement in reducing consumption rates? (4) How do family dynamics relate to risk?

E. Prior Art & Challenges

There has been prior work done on substance abuse and parental associations in Chile, observing how it has impacted the academic performances of Chilean adolescents, or the probability of engaging in high-risk behavior [4]. In some of these studies, the research was conducted through questionnaires in state-funded schools in the capital, Santiago de Chile, and the findings were subsequently analyzed

to assess parental attitudes toward substance abuse. In addition, the focus of the studies was on tobacco, alcohol, or marijuana separately. For this research, I will be generating a predictive model that determines the role of the family in the probability of a Chilean adolescent abusing substances, noting all three major drugs, for the general population of the country.

Some of the challenges presented in this research were the dataset responses. Some provinces received few responses, such as Arica, due to limited accessibility for adolescents compared with those living in Santiago, reflecting the area’s demographic makeup and differences in urban infrastructure. Furthermore, two social uprisings in 2011 and 2019 disrupted schools nationwide [5, 6]. With both protests dealing with inequality and rising costs in the country, it disrupted response rates for surveys being conducted, affecting the findings and predicaments of the factors.

II. DATA SOURCE AND RESEARCH APPROACH

A. Dataset

The data source is a survey conducted by the Servicio Nacional para la Prevención y Rehabilitación del Consumo de Drogas y Alcohol (SENDA), a department under the Interior Ministry of the Chilean government, named the Estudio Nacional de Drogas en Población Escolar (ENPE) [7]. The ENPE is conducted every two years for students to discuss their relationship with their parents, any history of drug use, and their performance in school [7]. In the ENPE responses database, 12 sets of responses were acquired from the years 2001 to 2023.

Each dataset contains roughly around 55,000 to 60,000 observations, with the 2011 and 2019 datasets having 33,509 and 46,628 observations respectively, due to the social protests that occurred in Chile in those years.

After the 12 datasets were gathered, I merged the datasets to begin the data exploration process. I will focus on the high school population, so I minimized the total amount of observations to just high school ages (13 to 18 years of age.) Table 1 presents the high school population sample size per year of the dataset, as well as the percentage of the sample size compared to the total sample size.

Further into the analysis, I will be dividing the sample size based on demographics, such as religion, region, and indigenous tribe affiliation.

B. Methodology & Tools

I will conduct this research in three phases, as shown in Fig. 1. In Phase 1, which is the current phase I am in, I am reviewing the ENPE survey questions to comprehend the social attitudes of Chilean youth towards drugs. This also involves doing a literature review on Chilean society and societal attitudes towards drug use. These questions are also interpreting how the family relationships are at home

TABLE I
ENPE HIGH SCHOOL POPULATION SAMPLE SIZE PER YEAR

Year	High School Pop. N	Sample Percentage
2001	56,784	96.7%
2003	57,020	97.5%
2005	59,038	98.6%
2007	50,252	98.5%
2009	46,892	95.7%
2011	31,297	98.0%
2013	55,771	97.4%
2015	52,864	97.1%
2017	53,462	97.7%
2019	44,549	95.5%
2021	57,858	97.9%
2023	47,759	97.0%

Note: The percentages represent the total percentage of the individual dataset, not the total merged dataset.

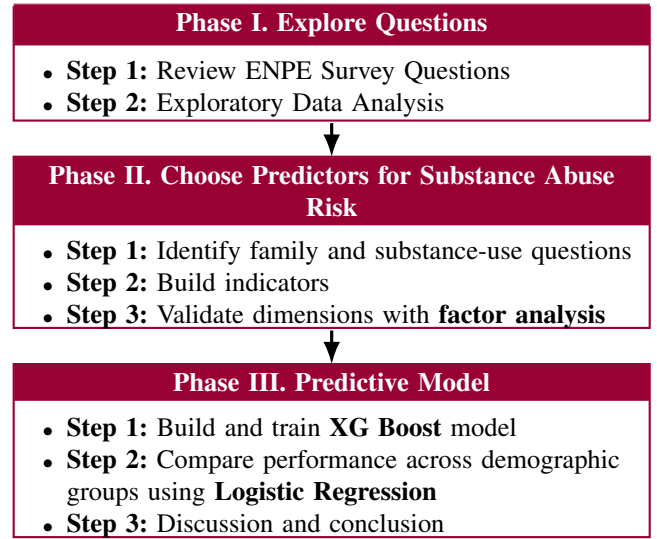


Fig. 1. Methodology Approach

and how they are doing in school, both academically and socially. With the review, I conduct summary statistics on the demographics presented in the dataset and an exploratory data analysis (EDA) on some key questions. In Phase 2, I will be choosing risk predictors for the analysis. First, I will identify family and substance-use questions that are essential for building indicators. The indicators are to examine the extent of family relationships and drug use among Chilean adolescents. From there, I will perform a factor analysis to validate dimensions and ensure the predictors are processed before generating the predictive model. As for the final phase, Phase 3, I will build and train an XG Boost model to classify and understand the contribution to each predictor chosen from Phase 2 and observe any relationships seen throughout the model. Moreover, I will compare performances across distinct demographic groups using logistic regression to analyze the correlations and likelihood of the predictors belonging to a contribution of substance abuse among the Chilean youth population. All three phases will be done using RStudio.

III. PRELIMINARY RESULTS

A. Demographics Summary Statistics

Figure 2 displays where the responses for the ENPE are coming from. In the figure, it shows that there are some regions with no answers, but there have been new regions that are providing responses, such as in the northern part of the country (Arica). The Santiago Metropolitan Area accounts for the majority of responses, providing 37.3% of the responses throughout 2001 to 2023.

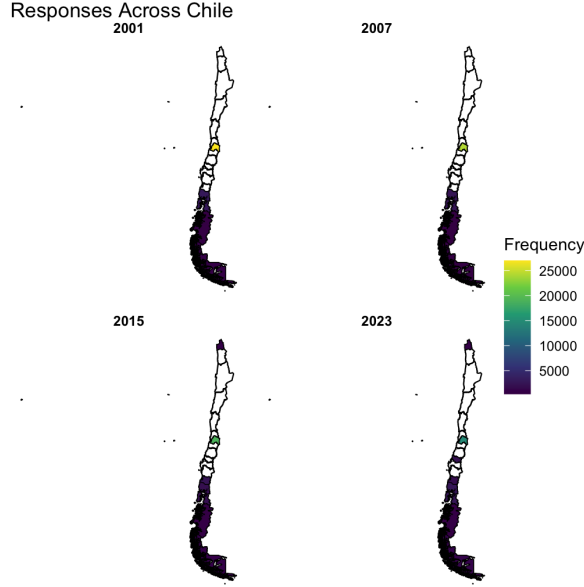


Fig. 2. Map Representation of ENPE Responses Across Chile

This also reflects upon the changes in the survey questions, as there has been more questions about indigenous ties and specificity of area in the region selected. For example, with the majority of the responses being from the Santiago Metropolitan Area, there has been more questions about where the adolescents reside in the capital.

Figure 3 demonstrates the distribution of the high school population in Chile across the years. The distribution across the years has been uniform, through standardization, illustrating how the population is appropriate for the size and study of this research. In addition, as mentioned before when discussing the data source, all of the datasets have around 90% or more of the population being in high school. As for the general dataset, the median age for the survey is 15 years of age, with 10 and 21 years of age being the range of the datasets. There is a value of 99 in the dataset, which indicates that there was no response of the individual when confirming their age.

Figure 4 illustrates the gender distribution across the years in the ENPE surveys. The distribution has been shifting to more male responses across the years, although it has been almost equal throughout the years. This is important to

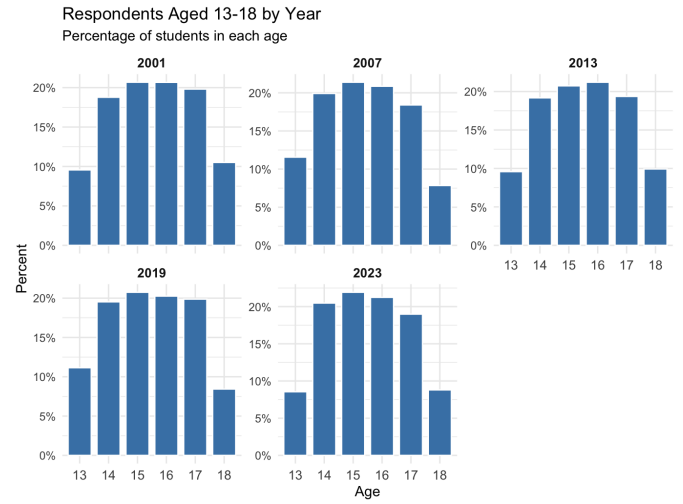


Fig. 3. Distribution of Age Percentages Across the High School Population

consider as gender can play a role in substance use, alongside how gender impacts the family dynamic in a household.

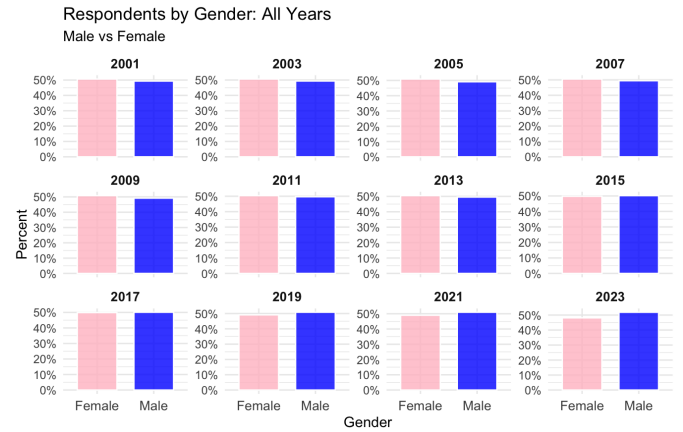


Fig. 4. Distribution of Gender Percentages Across the High School Population

B. Exploratory Data Analysis

In this current stage of the research, I am exploring the use of substances among the Chilean high school population. Figure 4 illustrates the trend across Chilean adolescents who have tried alcohol before. Figures 5 and 6 illustrate the trends across trying tobacco and marijuana, respectively.

Alcohol, out of the three drugs explored, has the highest percentage of adolescents being exposed to it, with the line remaining around the 70% mark throughout the years. This is based on how Chilean society approaches alcohol, as it is deemed a "softer" drug that is socially acceptable to try at a younger age [8]. This is contrast to tobacco exposure, as Chilean adolescents had a high percentage of exposure in 2001, but was steadily decreasing afterwards. As for marijuana, the drug is has the lowest percentage of exposure

among adolescents, with an increase from 2011 to 2015, but decreasing afterwards. A reason for the low percentages can be based on the legislation regarding cannabis use in Chile, as it is not fully legalized for recreational use [9].

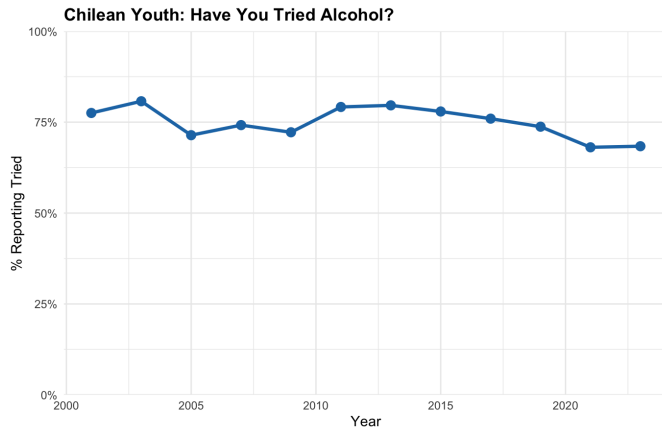


Fig. 5. Trend Line of Chilean Adolescents Trying Alcohol

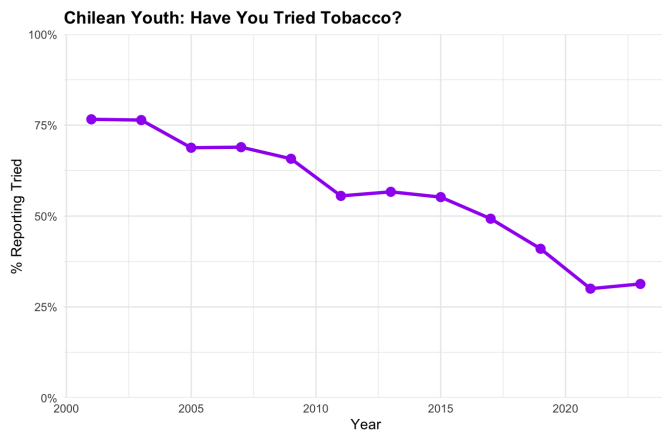


Fig. 6. Trend Line of Chilean Adolescents Trying Tobacco

Another observation to keep aware of was how parents approach schooling for their children. One of the main questions in the surveys was "How attentive are your parents when it comes to your schooling?", a beginner question to how academics and family structure can be seen when it comes to drug use. Figure 8 demonstrates bar plots across the years about schooling. The lowest level of attention is "Nothing" and the highest level of attention is "A lot", which there has decreased throughout the years. Nonetheless, parents are still being responsible for their children attending school, as "Quite" has been a frequent answer across the years.

IV. DISCUSSION AND CONCLUSION

A. Discussion

During the explanatory data analysis, I mainly focus on how drugs are being exposed and the family structures back at home. While Figures 5, 6, and 7 are showing the main

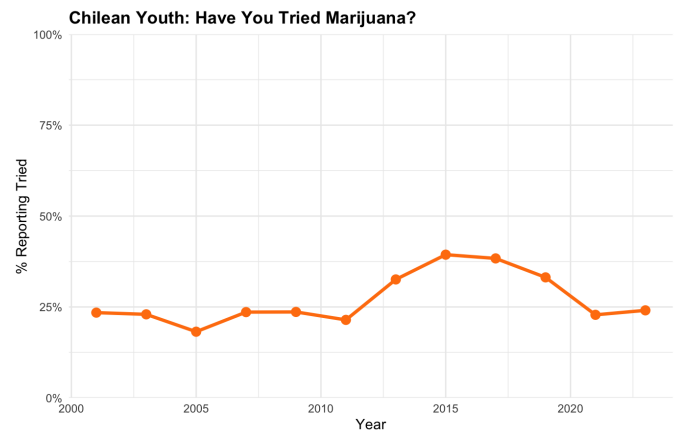


Fig. 7. Trend Line of Chilean Adolescents Trying Marijuana

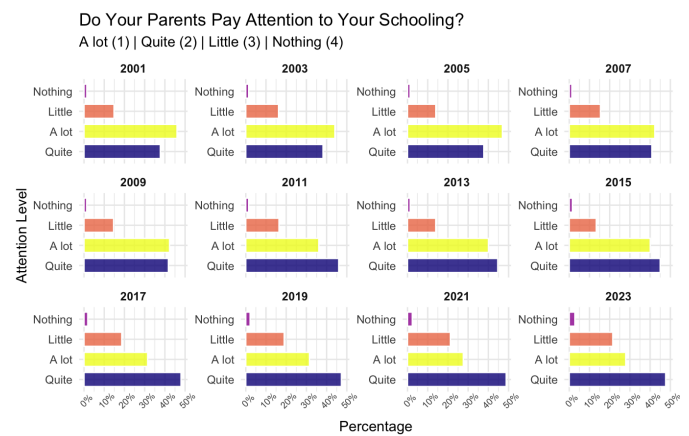


Fig. 8. Bar Plots Across the Years About Schooling

drugs that Chilean adolescents are trying, it is important to note that other drugs are being tried out, according to the survey. These drugs are part of another drug family, such as benzodiazepines, or a different type of method to consume, such as cocaine extract (or "pasta base" in Spanish). This will be further investigated as I get ahead in the research.

As for family structures, Figure 8 discusses how families approach society for their children. This question leads to more questions about how children feel about their parents and their approach to socialization, as there are questions relating to how parents view their friends, and how parents should dictate their use of drug consumption. A consideration for this research is to understand how the Chilean school system works; post-Pinochet, the system was mostly privatized and is still being reformed to provide for Chilean families who do not have the means to send their children to private school [9]. This will be further investigated as I explore the different types of schools seen in Chile, as well as the neighborhoods in which families reside. Another consideration for this research is seeing a specific region, such as the Santiago Metropolitan Area, and exploring the neighborhood context alongside the schools that

Santiago residents send their children to [3]. This consideration will be interesting to explore because, despite the regions providing enough information about the schools, having more specificity can provide more observations on how family dynamics can be related across different neighborhoods. Aside from Santiago, there is a probability that I can explore another region, such as Valparaíso or Antofagasta, two regions that can rank high in ENPE responses nationally.

B. Limitations

There were several limitations for this research, with a probability of more happening as I get further along. As mentioned before, some of the datasets were disrupted due to social uprisings; Figure 9 demonstrates that the 2011 and 2021 datasets have the highest percentages of missing data, with 25.5% & 9.28% of the data missing values. The social uprisings for both years were related to school reforms, due to government policies impacting the cost of education or changing the system that led to major protests throughout the country [5]. The 2021 dataset has been impacted due to the COVID-19 pandemic, which was another reason for the social protests that occurred that same year [6]. The impact of missing values will be seen in not just in the exploratory data analysis, but also in creating the predictive models. Thus, this can create a skewed response towards certain factors that must be addressed through normalization to ensure accurate predictions. In addition, the 2025 ENPE dataset has not been released yet, and it has not been confirmed as the Chilean government changed from liberal to conservative in the 2025 general elections [10]. The change of government may or may not have an impact on releasing the 2025 dataset, as well as the adjustments to the survey questions.

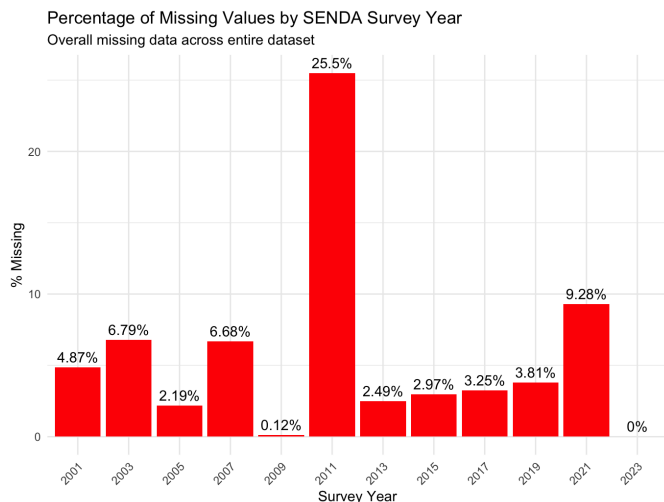


Fig. 9. Percentage of Missing Data Across All Datasets

C. Conclusion & Future Work

For the preliminary results, the drug use has been decreasing for alcohol, tobacco, and marijuana; there were slight jumps in 2023, but it has been minuscule. As for responses, there

has been a slight shift in gender and regions, seeing how the turnout has been expanding across the country. This is important to note as ENPE surveys are anonymous and do not reveal information about the student. With these first observations of the exploratory data analysis, there will be more questions explored about the use of other drugs, family relationships, and academic performance.

With my mentor, I will be coordinating meetings to review my progress throughout the research and discussing the next steps to prepare before entering the next two phases. This will be essential towards making sure the data has been well-explored, and questions have been examined thoroughly before choosing predictors for future analysis.

As for upcoming work, I will be exploring more demographic groups across the country, such as the type of schools Chilean adolescents attend and the indigenous population in the surveys. This will be essential towards providing more factors for predictive modeling and associations. Some other considerations for future work are seeing the differences in family structures across the regions, observing a difference in what drug is seen more in certain regions, investigating changes in society post-Pinochet and how Pinochetism impacts dynamics in present-day Chile, and perhaps having a comparison with another Latin American country, such as Colombia or Brazil. With these considerations, it is vital to note that there will be articles reviewed in order to understand Chilean society and its societal attitudes regarding upbringing and drugs in the country. Referring back to my second master's program, this research will be useful in investigating the Latin American region and highlighting the attitudes of substance abuse across different countries.

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